Fish sauce (fermented anchovy/marine sauce): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified standard for Fish sauce (fermented anchovy/marine sauce) (Condiments, Sauces and Dressings), stated as consumed. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; face value, no reconstitution; read-across / control / not-material for unregulated cells; no occurrence percentiles)

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	300	human-grade read-across (European Union)
Cadmium (Cd)	Tier 1	50	human-grade read-across (CR)

Total arsenic (tAs)	Tier 2	2,000	human-grade read-across (Australia / New Zealand)
Inorganic arsenic (iAs)	Tier 1	100	human-grade read-across (European Union)
Mercury, total (tHg)	Tier 1	300	human-grade read-across (European Union)
Methylmercury (MeHg)	Tier 1	300	bounded by tHg (reflex speciation)
Nickel (Ni)	Tier 2	20	human-grade read-across (European Union)
Tin (Sn)	Tier 2	100,000	human-grade read-across (European Union)
Aluminium (Al)	Tier 2	200	human-grade read-across (European Union)
Total chromium (Cr)	Tier 2	50	human-grade read-across (AU)
Hexavalent chromium (Cr-VI)	Tier 2	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	300	European Union — human-food analog
Cadmium (Cd)	50	CR — human-food analog
Total arsenic (tAs)	2,000	Australia / New Zealand — human-food analog
Inorganic arsenic (iAs)	100	European Union — human-food analog
Mercury, total (tHg)	300	European Union — human-food analog

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Methylmercury (MeHg)	300	Methylmercury is a component of total mercury (MeHg <= tHg). The MeHg value (500 ug/kg) is not below the binding total mercury ceiling (300 ug/kg), so methylmercury is governed by and bounded by the total mercury ceiling; no separate MeHg ceiling is published. It is enforced by reflex speciation whenever tHg is elevated.
Nickel (Ni)	20	European Union — human-food analog
Tin (Sn)	100,000	European Union — human-food analog
Aluminium (Al)	200	European Union — human-food analog
Total chromium (Cr)	50	AU — human-food analog
Hexavalent chromium (Cr-VI)	≤ Cr	Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell.

4 | Cite this standard

Pendergrass, K. (2026). *Fish sauce (fermented anchovy/marine sauce): HMTc standard*. Institute of Contaminant Standards.

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References

[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Australia / New Zealand — human-food maximum levels
Binds tAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] CR — human-food maximum levels
Binds Cd. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] European Union — human-food maximum levels
Binds Pb, iAs, tHg, Ni, Sn, Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/fish-sauce. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/fish-sauce.